

H25 Correlation Breakdown Short

Shorting NQ when it breaks away to the downside from ES: research report with fixed-risk sizing (cohort member #21, C2845a074bb5a5501)

Prepared for the investment committee · 23 September 2026 · data through 10 August 2026 · all figures per 1 NQ contract (\$20/point) unless stated

Recommendation: THIRD CANDIDATE, LOWER CONVICTION. Add it last, at small risk, as a third sleeve next to H41 and H47. Do not trade it alone. It was profitable in every research stage, and both post-freeze tests were classified SUPPORTED (holdout +\$17.6k, PF 1.47; 2026 +\$15.6k, PF 1.93). Its daily P&L is essentially uncorrelated with H41 and H47. But the untouched holdout was only marginal (t 1.19, null p 0.059). It needs its exact stop/exit plan, it wins only 19% of the time, and a handful of trades carry the result. It is a clear step down from the first two candidates.

Executive summary

- **What it is.** NQ and ES normally move almost in lockstep. It sells NQ short when, over the last 30 minutes, that link has loosened (correlation ≤ 0.65) and NQ has fallen at least 2 standard deviations more than its usual relationship with ES implies, which points to Nasdaq-specific selling. The stop is 1× ATR. 25% comes off at +1R, and the rest runs to 16:00. Short only. **About 47 trades a year after the freeze, roughly one a week, often in clusters. Three-quarters are in the first hour.**
- **Evidence ladder.** Discovery 2010-18: +\$7.2k, 473 trades, PF 1.22, t 1.13 (weak). Confirmation 2019-22: +\$57.1k, 255 trades, PF 1.99, t 2.85, null p 0.020, all four years positive, family-wise p 0.099 (one of the few to survive search correction at 10%), tier PROMISING. **Holdout 2023-25: +\$17.6k, 137 trades, PF 1.47, t 1.19, null p 0.059, SUPPORTED.** 2026 audit: +\$15.6k, 32 trades, PF 1.93, SUPPORTED.
- **After the freeze** (2023-Aug 2026): 169 trades, **PF 1.61**, +\$33.2k on 1 NQ, +65R, +0.38R per trade, t 1.66 in R (suggestive, not conclusive). Under stress costs it is +\$31.1k (PF 1.56).
- **How it makes money.** Only 19% of trades are winners. 80% are stopped out, usually within about 8 minutes, averaging -0.92R. The 20% still open at 16:00 average +5.3R. The average winner is 6.6× the average loser.
- **Main weaknesses.** (1) The holdout, the cleanest test, was marginal: t 1.19, and weaker than confirmation. (2) It needs its exact stop/exit plan: only 13 of 29 alternative exit schemes were profitable in the holdout, and most fixed-target schemes lost. (3) The 10 best post-freeze trades made 166% of net profit. (4) Discovery was weak: a 75R drawdown lasting almost six years.
- **Fixed risk.** Works from \$150 up with nothing skipped, and the profit factor is almost unchanged (about 1.55). Alone it is too slow and fragile for a prop evaluation. As a third sleeve next to H41 and H47 it adds a few points of pass rate and shortens the time to pass. Keep it at \$100-\$150.

Headline results by research stage

Period	Trades	Per week	Net P&L	PF	Win %	Net R	Avg R	Max DD	Verdict
Discovery 2010-18	473	1.1	\$7,228	1.22	17.1%	+51.5	+0.109	\$5,889	PRIMARY
Confirmation 2019-22	255	1.2	\$57,109	1.99	21.2%	+149.8	+0.587	\$6,058	PROMISING
Holdout 2023-25	137	0.9	\$17,570	1.47	19.0%	+36.4	+0.266	\$6,188	SUPPORTED
2026 audit (Jan-Aug)	32	1.0	\$15,603	1.93	21.9%	+28.5	+0.889	\$5,688	SUPPORTED
2024-2025 only	84	0.8	\$19,279	1.84	19.0%	+37.6	+0.448	\$5,726	-
All post-freeze 2023-26	169	0.9	\$33,173	1.61	19.5%	+64.9	+0.384	\$6,188	-

Stages before the freeze (discovery, confirmation) were used to select the strategy; 2023 onward was never seen before the cohort was frozen on 22 Sep 2026 23:21 UTC. Verdicts are the campaign's preregistered classifications. R = P&L divided by the trade's own initial risk.

1. How this strategy was found and tested

The strategy came out of the QuantLab V3 research campaign: 53 preregistered hypothesis families (33,077 specifications) were searched on 2010–2018 NQ/ES 1-minute data. Survivors were frozen and re-tested unchanged on 2019–2022 against 100 synthetic ‘no-edge’ market worlds, clustered into distinct behaviours, and 25 were selected into a frozen cohort. The cohort was then run once on the untouched 2023–2025 holdout, and once on 2026 data to 10 Aug 2026. Nothing was changed after any later stage was seen; that is what makes the post-2022 numbers honest out-of-sample evidence.

Stage	Dates	Role	What could change
Discovery	2010-06-07 → 2018-12-31	search (in-sample)	everything: rules, grid, management
Confirmation	2019 → 2022	out-of-sample test vs null worlds	only which frozen candidates are kept
Final cohort freeze	22 Sep 2026 23:21 UTC	spec hash f5666dac...	nothing after this point
Holdout	2023 → 2025	untouched out-of-sample	nothing (report only)
2026 audit	2026-01-01 → 2026-08-10	secondary, low power	nothing (report only)
True forward	from 2026-09-24	live-market paper record	nothing (forward/run_forward.py)

Reproducibility. For this report every trade from 2010 to 2026 was regenerated with the frozen specification, the stage-gated data loader and the campaign engine. Trade counts and net P&L match the sealed campaign records exactly in all four periods for every strategy covered here.

Why R-multiples matter here. NQ traded near 2,000 in 2010 and near 27,000 in 2026, and the stop is volatility-based, so one trade's dollar risk ranged from roughly \$50 to over \$2,000 on one contract. Dollar totals are therefore dominated by recent, high-volatility years, and a single contract is a very different bet in 2012 than in 2026. We report both dollars (the campaign's preregistered metric) and R ($\text{P\&L} \div \text{initial risk}$), which approximates a book that risks a constant amount per trade. **The official verdicts use dollars; the R view is a descriptive, post-hoc lens and cannot overturn them.**

Hypothesis family

Preregistered hypothesis (H25). Over the last 30 minutes, the NQ/ES return correlation is at most cth , and NQ's beta residual (NQ's move minus what ES's move predicts) is at least rz standard deviations. Two opposite ideas were tested separately: continuation (follow NQ's break-away) and convergence (fade it). Grid: $\text{cth} \in \{0.5, 0.65\}$, $\text{rz} \in \{1, 2\}$, $\text{mode} \in \{\text{continuation, reversal}\}$, long / short / both. In discovery 564 specifications were tested, 40 were PRIMARY, and the best t was 2.16.

Rationale. NQ and ES share most of their constituents and normally correlate above 0.9. When NQ suddenly falls much more than ES while the usual link loosens, the selling is specific to the Nasdaq names: a mega-cap tech disappointment, sector rotation, or funds cutting tech exposure. That kind of de-risking is usually worked through the whole session, so the relative weakness tends to continue rather than snap back. The data agree on direction and asymmetry. Fading the break-away (reversal-short) lost money in most variants and periods. Following NQ-specific *rallies* (the long side) was roughly break-even.

Behaviour cluster. #14: PROMISING, 7 confirmation survivors, 2 entry rules. Its nearest cohort sibling is cohort member #14 (H25 both directions, holdout correlation 0.33), which was WEAKER_THAN_EXPECTED in the holdout. The short side is where this family's edge lives.

2. H25 (#21, short only): specification

Field	Frozen value
Strategy ID	C2845a074bb5a5501
Entry rule ID	E7cd08682905f49b9
Family / version	v3_h25_corr_breakdown (v3.0)
Instrument	NQ (E-mini Nasdaq-100 futures, \$20 per point), front contract, 1-minute bars
Direction	short
Mode	continuation
Long trigger	not traded
Short trigger	30-minute NQ/ES correlation ≤ 0.65 and NQ's 30-minute beta residual ≤ -2.0 sigma
Parameters	{"cth": 0.65, "mode": "continuation", "rz": 2.0}
Signal timing	evaluated at the close of each 1-minute bar
Entry order	market order at the open of the next 1-minute bar (same session and contract)
Entry window	09:30-15:30 New York; one position at a time; no pyramiding
Initial stop	1.0 $\times$ ATRh from entry (ATRh = ATR(14) of completed 15-minute bars); stop order, gaps through the stop fill at the bar open; rounded away from entry to the tick
Target	25% at +1R + 75% runs to the time exit
Break-even / trailing / partials	none / none / 25% / 75%
Time exit	flat at the close of the last bar before 16:00 New York
Same-bar ambiguity	pessimistic (stop assumed first)
Costs (baseline)	\$14 per round trip = \$4 commission + 1 tick slippage per side; also scored at \$0 / \$9 / \$26
Cost file version	costs_v2_approved_2026-09-21#a27ed5df0c04

Statistical evidence trail

Stage	Test / statistic	Result	Verdict
Discovery 2010-18	label (trades ≥ 30 , net > 0 , t ≥ 1)	473 trades, PF 1.22, t 1.13	PRIMARY
Discovery	parameter robustness (share of grid neighbours positive)	1.00	
Discovery	sub-periods positive (of 3) / years positive (of 9)	2 / 6	
Confirmation 2019-22	t-stat of mean trade	2.85	
Confirmation	null p (zero-drift worlds) / (drift worlds) / max	0.020 / 0.020 / 0.020	$p \leq 0.05$
Confirmation	family-wise p (max-t in same frequency bucket)	0.099	significant at 10%
Confirmation	parameter support / management support	1.00 / 0.15	
Confirmation	years positive (of 4); stress net	4; \$54,049	
Confirmation	evidence tier (preregistered)		PROMISING
Behaviour cluster	cluster at p 0.50 (members / entry rules)	#14 (7 / 2)	PROMISING
Holdout 2023-25	t-stat; z vs confirmation mean trade	1.19; -0.89	
Holdout	null p (100 holdout null worlds)	0.059	
Holdout	classification (preregistered)		SUPPORTED
2026 audit	t-stat; net	1.08; \$15,603	SUPPORTED

Null worlds randomly flip the sign of each minute's moves (keeping volatility, seasonality, gaps and the NQ-ES relation) so any directional edge is destroyed; p is the share of such worlds that match the real t-stat. The family-wise p corrects for having searched many candidates of similar frequency.

H25 (#21, short only): how it trades

Signal (checked at every 1-minute bar close, 09:30-15:30 New York). Two conditions must hold. (1) The correlation of NQ and ES 1-minute returns over the last 30 minutes is at most 0.65. (2) NQ's 30-minute move is at least 2 standard deviations below what its normal beta to ES predicts. When both hold, it sells NQ at market on the next bar. It holds one position at a time and can fire again later the same day. 75% of all trades start between 09:30 and 10:00, because the open is when Nasdaq-specific news hits.

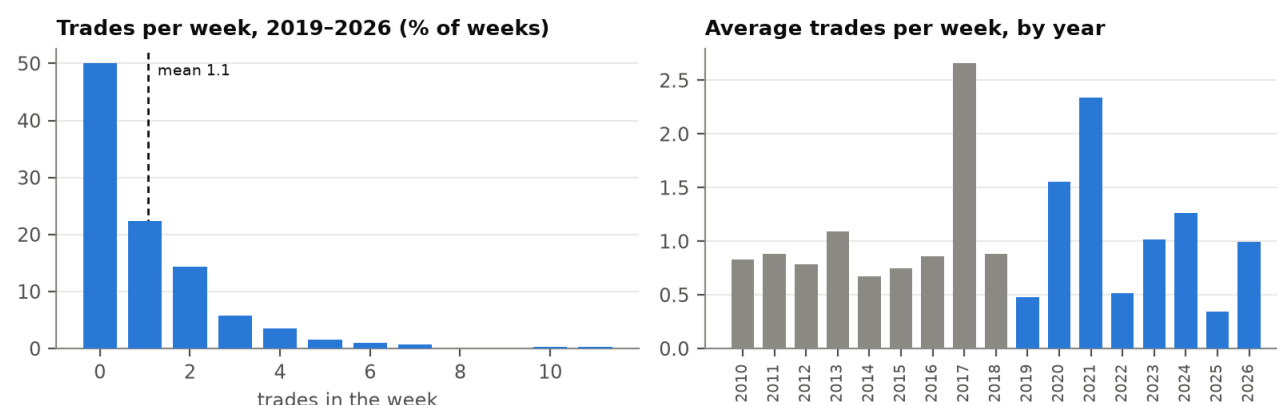
Risk and exit. The stop sits $1 \times \text{ATR}_h$ above the entry ($\text{ATR}(14)$ of completed 15-minute bars): about 18–21 NQ points in 2023–25 and 43 points in 2026. One NQ contract risked about \$825 in 2026; one MNQ about \$85. A quarter of the position is covered at $+1R$. The rest has no target, no trailing stop and no move to break-even, and is covered at the last bar before 16:00.

What a trade looks like. Most attempts fail almost immediately: the median stopped trade lasts 8 minutes. About 53% reach $+1R$ and bank the partial. The one trade in five that works turns into a trend-down day and is held for hours (median 380 minutes). This is descriptive only, and the rules are not changed because of it.



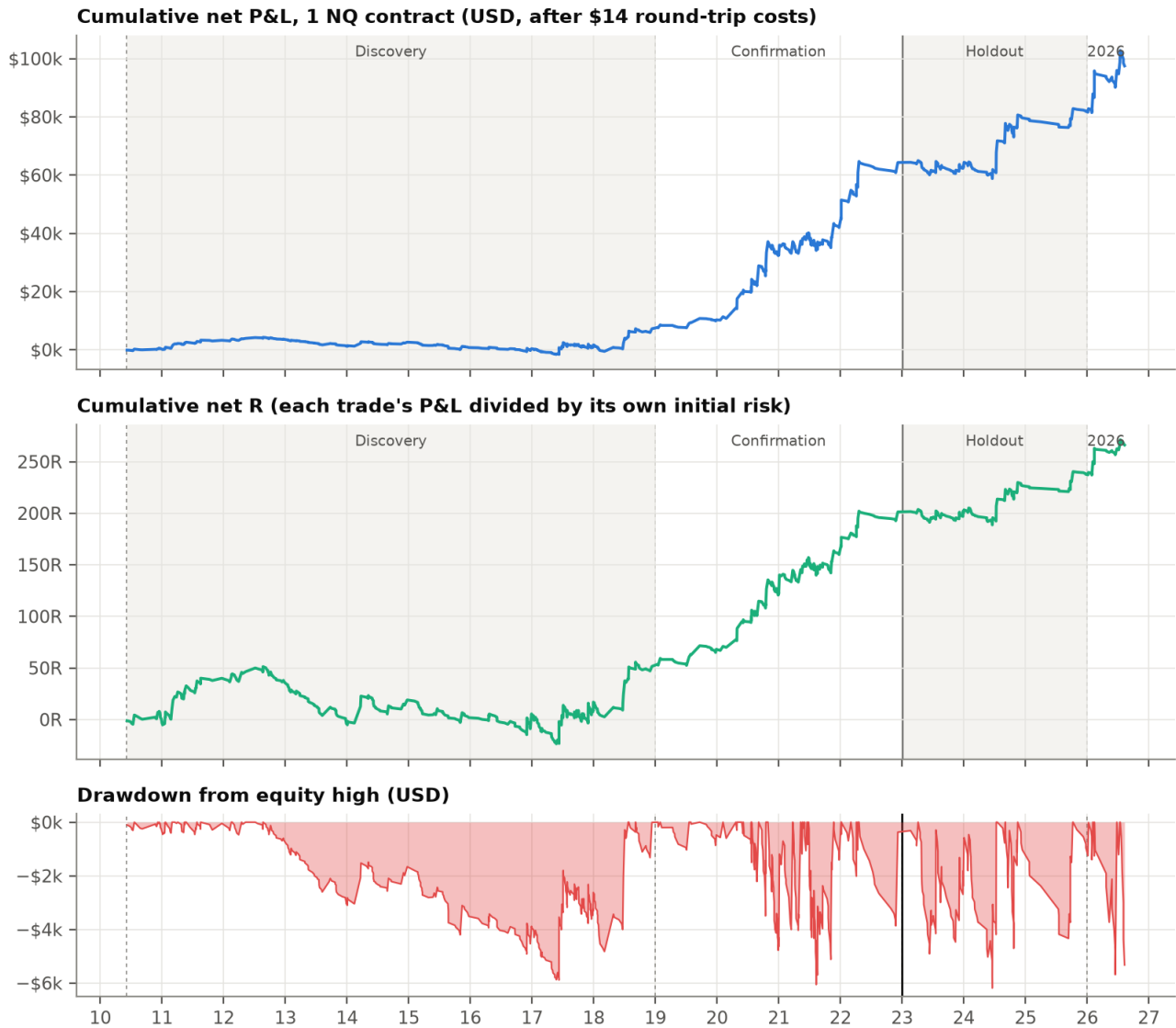
Real holdout trades, 1-minute NQ bars (grey band = bar range, line = close). Blue dotted = entry price, red dashed = the fixed stop; nothing else is managed.

How often it trades



Frequency (2019-Aug 2026)	Value
Average trades per week	1.1
Median trades per week	0
Weeks with no trade	50.1%
Weeks with exactly 1 trade	22.4%
Weeks with 2 or more trades	27.5%
Busiest week	11 trades
Longest gap without a trade	14 weeks
Average trades per year	56

H25 (#21, short only): performance



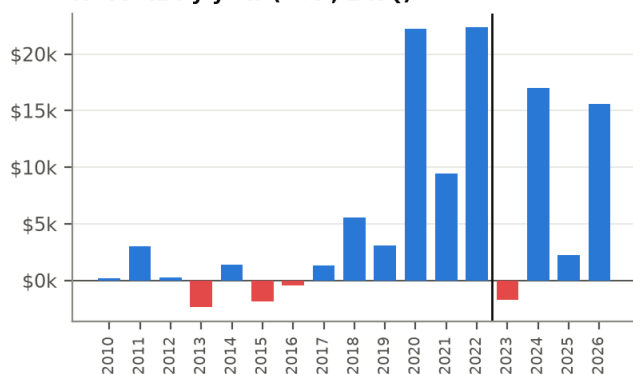
H25 (#21, short only): full statistics by period

Metric	Discovery 2010-18	Confirm. 2019-22	Holdout 2023-25	2026 Jan-Aug	2024-25	Post-freeze 2023-26	Full 2010-26
Trades	473	255	137	32	84	169	897
Trades / year	55	64	46	52	42	47	55
Trades / week (all weeks)	1.1	1.2	0.9	1.0	0.8	0.9	1.1
Weeks with at least 1 trade	54%	52%	43%	65%	40%	47%	52%
Trades / week (active weeks)	2.0	2.3	2.0	1.5	2.0	1.9	2.0
Days with a trade (of sessions)	17%	19%	14%	17%	13%	15%	17%
Max trades in one day	4	5	4	2	4	4	5
Long share	0%	0%	0%	0%	0%	0%	0%
Net P&L (baseline \$14)	\$7,228	\$57,109	\$17,570	\$15,603	\$19,279	\$33,173	\$97,510
Gross P&L (no costs)	\$13,850	\$60,679	\$19,488	\$16,051	\$20,455	\$35,539	\$110,068
Net P&L, moderate \$9	\$9,593	\$58,384	\$18,254	\$15,763	\$19,699	\$34,018	\$101,994
Net P&L, stress \$26	\$1,552	\$54,049	\$15,926	\$15,219	\$18,271	\$31,145	\$86,746
Profit factor (net)	1.22	1.99	1.47	1.93	1.84	1.61	1.68
Profit factor (gross)	1.51	2.11	1.54	1.98	1.93	1.68	1.82
Profit factor (stress)	1.04	1.90	1.41	1.89	1.77	1.56	1.57
Win rate	17.1%	21.2%	19.0%	21.9%	19.0%	19.5%	18.7%
Average trade	\$15	\$224	\$128	\$488	\$230	\$196	\$109
Average winner	\$494	\$2,122	\$2,115	\$4,625	\$2,637	\$2,647	\$1,440
Average loser	-\$84	-\$286	-\$337	-\$671	-\$337	-\$398	-\$198
Payoff ratio (avg W / avg L)	5.91	7.42	6.27	6.89	7.83	6.64	7.27
Expectancy (net R / trade)	+0.109	+0.587	+0.266	+0.889	+0.448	+0.384	+0.297
Total net R	+51.5	+149.8	+36.4	+28.5	+37.6	+64.9	+266.2
Median trade (R)	-0.90	-0.57	-0.57	-0.52	-0.55	-0.56	-0.78
t-stat of mean trade (USD)	1.13	2.85	1.19	1.08	1.43	1.61	3.30
t-stat of mean trade (R)	0.78	2.89	1.13	1.28	1.33	1.66	2.88
Sharpe, daily, annualised (USD)	0.38	1.41	0.68	1.36	1.00	0.84	0.81
Sharpe, daily, annualised (R)	0.26	1.42	0.65	1.62	0.94	0.87	0.70
Sortino (USD)	1.08	5.16	2.21	4.49	3.84	2.74	2.77
Max drawdown (USD)	\$5,889	\$6,058	\$6,188	\$5,688	\$5,726	\$6,188	\$6,188
Max drawdown (R)	74.9	17.2	16.4	6.3	16.4	16.4	74.9
Longest drawdown (calendar days)	2,149	230	462	124	319	462	2,149
Calmar (annual net / max DD)	0.14	2.36	0.95	4.45	1.68	1.48	0.97
Longest losing streak (trades)	23	14	14	6	12	14	23
Longest winning streak	3	3	2	2	2	2	3
Best trade	\$2,346	\$6,866	\$6,930	\$9,172	\$6,930	\$9,172	\$9,172
Worst trade	-\$434	-\$974	-\$869	-\$1,359	-\$869	-\$1,359	-\$1,359
Best day / worst day	\$2,248 / -\$708	\$6,866 / -\$1,358	\$6,930 / -\$1,317	\$8,548 / -\$2,453	\$6,930 / -\$1,317	\$8,548 / -\$2,453	\$8,548 / -\$2,453
Top-10 trades as % of net	211%	83%	228%	202%	196%	166%	64%
Positive months	39% of 95	49% of 43	39% of 28	43% of 7	41% of 17	40% of 35	42% of 173
Average hold (minutes)	86	104	84	108	95	89	92
Avg initial risk (NQ points)	4.4	17.3	19.7	41.3	20.7	23.8	11.7
Avg initial risk, 1 NQ (USD) = 1R	\$88	\$347	\$395	\$825	\$414	\$476	\$235
Average NQ price at entry	4,452	12,826	18,259	28,040	20,400	20,111	9,783

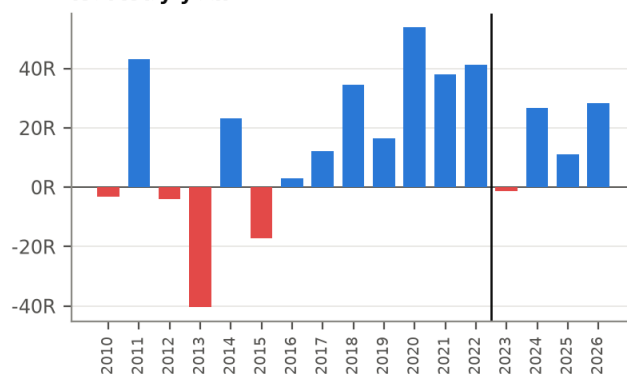
Sharpe and Sortino use every business day in the period (zero on no-trade days) and are scale-free. The strategy is always flat overnight.

H25 (#21, short only): calendar

Net P&L by year (USD, 1 NQ)



Net R by year

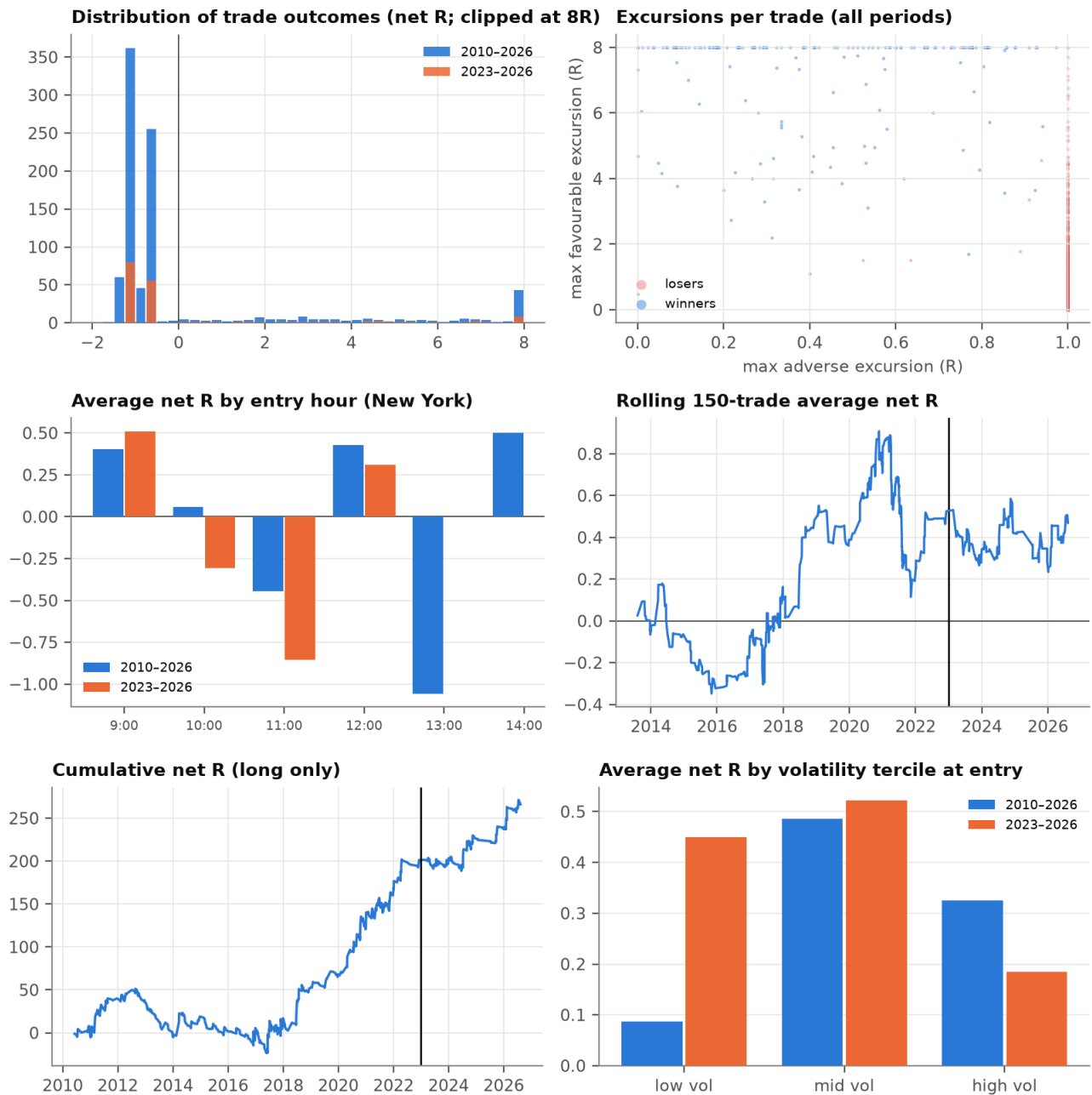


Year	Period	N	Net P&L	Gross	Stress	PF	Win %	Avg trade	Net R	Avg R	Max DD	1R (USD)
2010	Disc.	25	\$208	\$558	-\$92	1.20	16%	\$8	-3.2	-0.129	\$347	\$52
2011	Disc.	46	\$3,057	\$3,701	\$2,505	2.41	33%	\$66	+43.2	+0.940	\$456	\$69
2012	Disc.	41	\$298	\$872	-\$194	1.14	22%	\$7	-4.0	-0.097	\$715	\$66
2013	Disc.	57	-\$2,348	-\$1,550	-\$3,032	0.27	7%	-\$41	-40.3	-0.708	\$2,348	\$62
2014	Disc.	35	\$1,402	\$1,892	\$982	1.75	20%	\$40	+23.3	+0.666	\$1,081	\$68
2015	Disc.	39	-\$1,854	-\$1,308	-\$2,322	0.47	10%	-\$48	-17.2	-0.441	\$2,542	\$103
2016	Disc.	45	-\$418	\$212	-\$958	0.87	11%	-\$9	+3.0	+0.066	\$1,491	\$83
2017	Disc.	139	\$1,310	\$3,256	-\$358	1.12	17%	\$9	+12.1	+0.087	\$1,972	\$96
2018	Disc.	46	\$5,571	\$6,215	\$5,019	2.07	22%	\$121	+34.6	+0.753	\$2,204	\$165
2019	Conf.	25	\$3,061	\$3,411	\$2,761	2.43	28%	\$122	+16.6	+0.664	\$1,031	\$148
2020	Conf.	81	\$22,208	\$23,342	\$21,236	2.23	20%	\$274	+53.9	+0.665	\$4,780	\$356
2021	Conf.	122	\$9,473	\$11,181	\$8,009	1.33	20%	\$78	+38.0	+0.311	\$6,058	\$350
2022	Conf.	27	\$22,366	\$22,744	\$22,042	3.77	26%	\$828	+41.4	+1.532	\$3,866	\$488
2023	Holdout	53	-\$1,710	-\$968	-\$2,346	0.88	19%	-\$32	-1.2	-0.023	\$4,911	\$365
2024	Holdout	66	\$17,030	\$17,954	\$16,238	1.94	18%	\$258	+26.6	+0.404	\$5,726	\$406
2025	Holdout	18	\$2,249	\$2,501	\$2,033	1.47	22%	\$125	+11.0	+0.612	\$3,310	\$444
2026	Audit	32	\$15,603	\$16,051	\$15,219	1.93	22%	\$488	+28.5	+0.889	\$5,688	\$825

Monthly net R, 2019-2026 (blue = gain, red = loss; blank = no data)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2019	+7.8	-1.1		-2.4	-1.1		+7.9		+8.9	-0.6	-3.1	+0.3
2020	-1.2	+3.0		+18.4	+6.6	+0.3	+9.2	-3.6	+11.9	+22.8	-4.9	-8.6
2021	+18.9	-5.4	+6.1	-6.6	+13.2	+2.0	-1.1	-3.3	+5.4	-6.4	+18.3	-3.1
2022	+16.8	-1.6	+3.1	+22.8	-1.0	-1.0	-2.1	-1.0		-1.0	-2.1	+8.6
2023		+0.3	-1.6	-2.1	-3.7	+0.4	+6.5	-2.4	-1.6	-2.1	+0.4	+4.5
2024	+5.0	-6.3	-2.1		-4.8	+2.7	+19.1	-1.6	+11.2	-2.6	+9.0	-3.2
2025	-2.1		-0.5				-2.6		+9.1	+9.2		-2.1
2026	-0.7	+25.0		-2.1	-0.2	+2.7	+6.7	-3.0				

H25 (#21, short only): trade anatomy



Volatility tercile = initial stop distance as a share of price, ranked within each calendar year.

H25 (#21, short only): breakdowns (all periods vs post-freeze 2023-26)

Exit reason

Exit	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
16:00 flat	175	+5.311	1,156.23	\$241,732	33	+5.381	-	\$87,364
Stop-loss hit	722	-0.919	0.00	-\$144,223	136	-0.829	0.00	-\$54,192

Entry hour (New York)

Entry hour	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
09:00-09:59	674	+0.405	1.87	\$92,818	144	+0.511	1.87	\$38,573
10:00-10:59	154	+0.058	1.34	\$9,276	21	-0.305	0.54	-\$3,572
11:00-11:59	40	-0.444	0.32	-\$5,139	3	-0.854	0.00	-\$1,884
12:00-12:59	18	+0.428	2.02	\$1,409	1	+0.311	-	\$56
13:00-13:59	7	-1.056	0.00	-\$920	0	-	-	\$0
14:00-14:59	4	+0.501	1.25	\$65	0	-	-	\$0

Weekday

Weekday	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Mon	206	+0.006	0.87	-\$4,908	46	-0.539	0.31	-\$11,882
Tue	175	+0.547	2.40	\$36,304	35	+1.007	2.24	\$13,481
Wed	201	+0.303	1.68	\$22,401	35	+0.449	2.16	\$10,422
Thu	164	+0.159	2.04	\$30,755	34	+1.185	2.91	\$21,112
Fri	151	+0.545	1.68	\$12,957	19	-0.083	1.01	\$39

H25 (#21, short only): robustness

Parameter neighbourhood. Every grid point of the family with the same stop and exit, both directions separately. If the frozen point is a lucky outlier, its neighbours will look much worse. Rerun post hoc for this report; purely descriptive, and it does not re-select anything.

Entry parameters	Dir.	Disc. n / PF	Conf. n / PF	Holdout n / PF	Holdout net	2026 n / PF	2026 net
cth=0.5, rz=1.0, mode=continuation	long	459 / 1.02	217 / 1.16	91 / 0.94	-\$1,482	11 / 3.97	\$10,524
cth=0.5, rz=1.0, mode=continuation	short	394 / 1.09	201 / 1.49	69 / 0.98	-\$327	19 / 1.62	\$5,095
cth=0.5, rz=1.0, mode=continuation	both	803 / 1.05	389 / 1.42	153 / 0.92	-\$3,323	29 / 2.06	\$13,035
cth=0.5, rz=2.0, mode=continuation	long	180 / 1.20	127 / 1.11	67 / 0.99	-\$97	10 / 2.73	\$6,895
cth=0.5, rz=2.0, mode=continuation	short	165 / 1.86	113 / 1.47	49 / 1.34	\$3,816	14 / 0.83	-\$1,066
cth=0.5, rz=2.0, mode=continuation	both	343 / 1.53	234 / 1.33	115 / 1.12	\$3,499	24 / 1.56	\$5,829
cth=0.5, rz=1.0, mode=reversal	long	395 / 1.15	194 / 1.19	64 / 1.31	\$4,396	20 / 1.21	\$2,201
cth=0.5, rz=1.0, mode=reversal	short	512 / 0.79	236 / 0.91	85 / 0.45	-\$12,491	13 / 0.27	-\$5,151
cth=0.5, rz=1.0, mode=reversal	both	800 / 0.90	356 / 0.94	130 / 0.53	-\$16,310	29 / 0.84	-\$2,426
cth=0.5, rz=2.0, mode=reversal	long	178 / 1.31	118 / 1.22	43 / 0.78	-\$2,320	13 / 1.48	\$3,323
cth=0.5, rz=2.0, mode=reversal	short	193 / 0.76	133 / 0.62	64 / 0.57	-\$7,460	12 / 0.33	-\$4,690
cth=0.5, rz=2.0, mode=reversal	both	352 / 1.04	229 / 0.83	98 / 0.37	-\$16,853	25 / 0.90	-\$1,368
cth=0.65, rz=1.0, mode=continuation	long	1246 / 0.90	460 / 1.16	248 / 1.09	\$5,743	30 / 1.42	\$7,015
cth=0.65, rz=1.0, mode=continuation	short	1147 / 0.93	452 / 1.39	199 / 1.22	\$12,263	45 / 1.66	\$16,162
cth=0.65, rz=1.0, mode=continuation	both	2114 / 0.92	792 / 1.32	415 / 1.22	\$25,112	73 / 1.45	\$19,063
cth=0.65, rz=2.0, mode=continuation	long	551 / 0.91	292 / 1.03	194 / 1.05	\$2,852	22 / 1.96	\$9,817
cth=0.65, rz=2.0, mode=continuation ◀ frozen	short	473 / 1.22	255 / 1.99	137 / 1.47	\$17,570	32 / 1.93	\$15,603
cth=0.65, rz=2.0, mode=continuation	both	1006 / 1.09	529 / 1.49	324 / 1.24	\$22,709	54 / 1.94	\$25,420
cth=0.65, rz=1.0, mode=reversal	long	1150 / 0.99	426 / 1.11	198 / 1.13	\$6,917	41 / 1.15	\$3,565
cth=0.65, rz=1.0, mode=reversal	short	1452 / 0.69	521 / 0.94	250 / 0.56	-\$33,681	33 / 1.21	\$4,072
cth=0.65, rz=1.0, mode=reversal	both	2092 / 0.88	750 / 1.01	384 / 0.76	-\$27,097	63 / 0.73	-\$10,017
cth=0.65, rz=2.0, mode=reversal	long	511 / 1.08	266 / 0.91	133 / 1.11	\$4,008	30 / 0.77	-\$4,271
cth=0.65, rz=2.0, mode=reversal	short	596 / 0.64	314 / 0.76	185 / 0.87	-\$7,016	23 / 0.76	-\$3,426
cth=0.65, rz=2.0, mode=reversal	both	1032 / 0.78	517 / 0.85	290 / 0.87	-\$10,546	48 / 0.22	-\$23,870

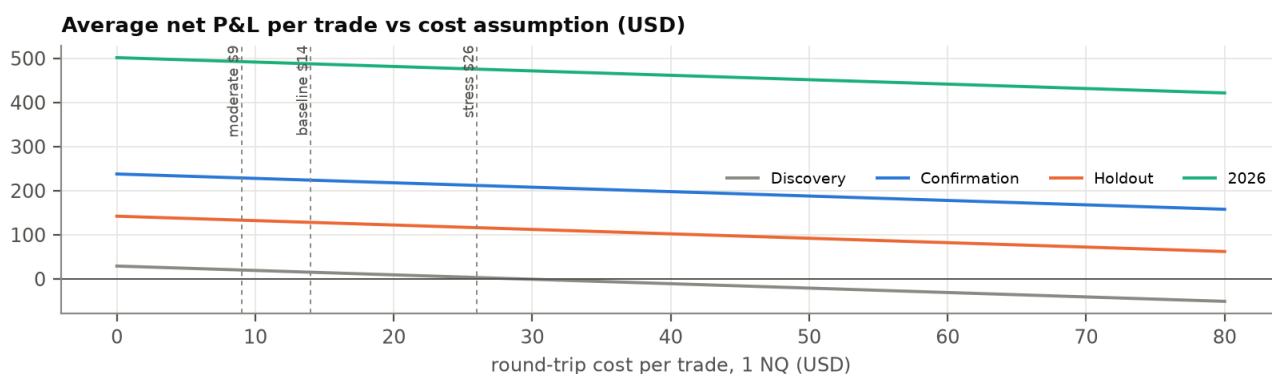
Neighbours with ≥ 20 trades and positive net: Discovery 12/24; Confirmation 16/24; Holdout 11/24; 2026 11/17

Stop / exit management. The frozen entry under every management variant from the campaign library (discovery stages A and B).

Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; single target at +1R	0.72	1.13	163 / 0.94	-\$1,892	1.37	\$4,325
1×ATR stop; single target at +1.5R	0.76	1.26	154 / 0.91	-\$3,336	0.97	-\$550
1×ATR stop; single target at +2R	0.77	1.12	146 / 0.95	-\$2,184	1.12	\$2,194
1×ATR stop; single target at +3R	0.82	1.23	140 / 1.11	\$4,590	1.27	\$5,037
1×ATR stop; single target at +4R	0.94	1.17	138 / 1.04	\$1,708	1.08	\$1,637
1×ATR stop; no target, flat 16:00	1.30	2.04	137 / 1.53	\$24,377	1.92	\$19,872
1×ATR stop; no target, exit after 60 min	1.17	1.46	137 / 0.96	-\$1,448	2.03	\$19,817
1.5×ATR stop; single target at +1R	0.78	1.29	141 / 0.97	-\$1,174	0.79	-\$4,911
1.5×ATR stop; single target at +2R	0.78	1.20	125 / 1.03	\$1,260	0.98	-\$509
1.5×ATR stop; no target, flat 16:00	1.18	1.80	121 / 1.22	\$12,676	1.32	\$10,061
5-min swing stop; single target at +1R	0.87	1.11	135 / 1.22	\$9,455	1.88	\$14,642
5-min swing stop; single target at +2R	1.01	1.04	127 / 0.99	-\$473	2.42	\$28,741
5-min swing stop; no target, flat 16:00	1.11	1.78	123 / 1.18	\$11,393	2.95	\$44,760
1×ATR stop; stop to entry at +0.5R, target +2R	0.39	0.34	166 / 0.26	-\$18,494	0.11	-\$7,864
1×ATR stop; stop to entry at +0.5R, no target	0.65	0.41	164 / 0.65	-\$8,621	0.85	-\$1,309
1×ATR stop; stop to entry at +1R, target +2R	0.61	0.93	155 / 0.66	-\$11,175	0.82	-\$2,185
1×ATR stop; stop to entry at +1R, no target	1.06	1.58	151 / 1.19	\$5,761	1.78	\$9,385
1×ATR stop; stop to entry at +1.5R, target +2R	0.69	1.13	150 / 0.93	-\$2,670	1.03	\$525
1×ATR stop; stop to entry at +1.5R, no target	1.25	1.81	143 / 1.32	\$11,123	2.11	\$19,118
1×ATR stop; no target, 1.5×ATR trailing stop	0.99	1.23	150 / 0.72	-\$9,255	1.97	\$12,318
1×ATR stop; no target, 2×ATR trailing stop	1.11	1.50	139 / 0.95	-\$1,801	1.56	\$9,157

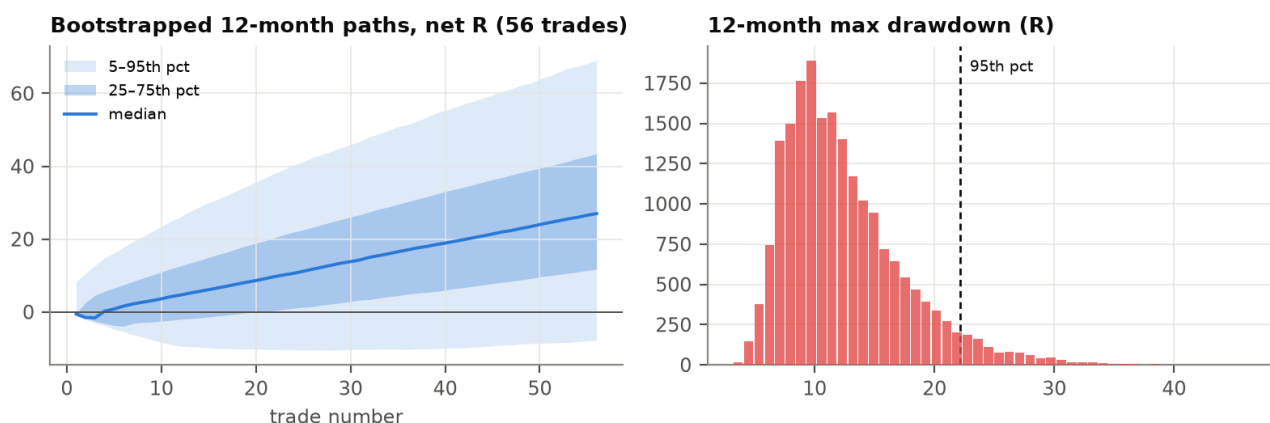
Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; no target, 3×ATR trailing stop	1.12	1.83	137 / 1.38	\$15,297	1.81	\$15,342
1×ATR stop; no target, trail beyond last 5-min swing	1.36	1.80	144 / 1.42	\$16,409	2.78	\$28,333
1×ATR stop; 50% at +1R, 50% at +2R	0.74	1.16	146 / 0.91	−\$2,952	1.25	\$2,984
1×ATR stop; 50% at +1R, 50% at +3R	0.75	1.23	140 / 1.00	\$10	1.33	\$3,917
1×ATR stop; 50% at +1R, rest trails 2×ATR	0.94	1.34	139 / 0.92	−\$2,241	1.51	\$5,977
1×ATR stop; 25% at +1R, 75% runs to 16:00 ◀ frozen	1.22	1.99	137 / 1.47	\$17,570	1.93	\$15,603
1×ATR stop; 50% at +1R then stop to entry, rest +3R	0.68	1.07	154 / 0.85	−\$4,814	1.17	\$1,968
1×ATR stop; stop to entry at +1R, 2×ATR trail	0.93	1.15	151 / 0.69	−\$9,399	0.68	−\$3,890

Cost sensitivity. The line shows what the average trade would be at any round-trip cost.



Break-even round-trip cost per contract (gross average trade): Discovery \$29; Confirmation \$238; Holdout \$142; 2026 \$502. The baseline assumption is \$14.

H25 (#21, short only): risk, sizing and forward expectations



Bootstrap of 424 trades from 2019–Aug 2026 (all out-of-sample relative to discovery), 56 trades per simulated year, 20,000 paths, trades drawn independently. This ignores regime persistence, so real drawdowns cluster more than shown. It assumes the 2019–26 edge continues unchanged; the section 'Will it perform like this in the future?' repeats it with smaller edges.

Forward 12-month statistic (bootstrap)	In R	1 NQ at 2026 risk (1R ≈ \$825)	1 MNQ (1R ≈ \$83)
Median 12-month result	+27.0	\$22,286	\$2,229
5th percentile (bad year)	−7.7	−\$6,378	−\$638
95th percentile (good year)	+69.1	\$57,017	\$5,702
Probability of a losing year	11%		
Median max drawdown	11.4	\$9,373	\$937
95th pct max drawdown	22.1	\$18,273	\$1,827
99th pct max drawdown	28.5	\$23,556	\$2,356
Historical max drawdown 2010–26	74.9		

Suggested capital rule of thumb (not a margin requirement): hold at least the 99th-percentile 12-month drawdown plus exchange margin per unit traded. Dollar outcomes on one full NQ contract are far larger today than in most of the backtest; micro contracts (MNQ, 1/10 size) allow risk to be held near-constant per trade.

Portfolio context (25-strategy frozen cohort, 1 contract each)

Period	Corr. with rest of cohort (daily)	Median pairwise corr.	Most correlated members	Share of cohort P&L
holdout	0.21	0.03	#14 H25 both (0.33), #9 H08 short (0.26), #7 H07 both (0.18), #2 H45 both (0.16)	3%
2026	0.10	−0.01	#6 H06 both (0.39), #15 H41 short (0.28), #14 H25 both (0.18), #1 H20 both (0.16)	16%

Strengths and weaknesses

Strengths

- **Positive in every stage, including both post-freeze tests.** Holdout +\$17.6k (PF 1.47) and 2026 +\$15.6k (PF 1.93) were both classified SUPPORTED. In R, 2024 (+26.6R), 2025 (+11.0R) and 2026 (+28.5R) were all positive.
- **One of the few to survive search correction.** The family-wise p of 0.099 in confirmation is much better than H41 (0.35) or H47 (0.64). Given how many specifications were searched, its confirmation result was unusual.
- **Clear direction, consistent with the idea.** All four short-continuation grid points were profitable in confirmation and three of four in the holdout. Fading the same signal (reversal-short) lost money in almost every variant and period.
- **A genuine diversifier, and short only.** Daily correlation after the freeze is about 0.14 with H41, about 0 with H47, and 0.21 with the rest of the cohort. It earns on Nasdaq-specific down days, which partly offsets the long-biased H47.
- **Costs are negligible.** The average trade is about \$200 on 1 NQ, and stress-cost PF after the freeze is 1.56.
- **Fixed-risk friendly.** It needs about 1 MNQ at \$100 even in 2026, and nothing is skipped from \$150 up.

Weaknesses and risks

- **The cleanest test was only marginal.** Holdout t 1.19, null p 0.059, and the holdout average trade was below confirmation (z -0.89). The post-freeze t is 1.66 in R. Positive, but it could still be luck.
- **It needs its exact exit plan.** Only 13 of 29 stop/exit variants were profitable in the holdout. 5 of 9 fixed-target variants lost, and so did every 'break-even, then target' variant. Only the runner-based exits worked (hold to close, wide trailing stops, and the frozen 25%/75% plan). The whole edge is in the few trades that trend all day, and any change that caps them removes it.
- **Very low win rate, extreme tail dependence.** Only 19% of trades win, and the 10 best post-freeze trades made 166% of net profit. Losing streaks of 14 trades happened in both confirmation and the holdout, and 23 in discovery.
- **Weak and long drawdown in discovery.** 2010-18 had PF 1.22 and t 1.13, with a 75R drawdown lasting 2,149 days. 2013 (-40R) and 2015 (-17R) were bad years. The strong results began around 2018-2020.
- **Clustered losses on a single morning.** Up to 4-5 entries can fire in one session, and the worst post-freeze day on 1 NQ was -\$2,453. This matters for daily loss limits.
- **Uneven activity.** Trades per year swing from 18 (2025) to 122 (2021) depending on how often NQ decouples from ES. Some years give it very few chances.

Will it perform like this in the future?

Short answer: possibly positive, and the least certain of the three. For persistence: a clear economic story (Nasdaq-specific de-risking persists), consistent direction across neighbours, search-corrected significance in confirmation, and both post-freeze periods profitable. For a large haircut: the holdout was only marginal and below confirmation, discovery was weak, and the result depends on a few large down days. The table resamples its own trades at five assumed average trades.

Scenario for the next 12 months	Avg R / trade	PF (R)	Median 12-mo	Bad year (5th pct)	Good year (95th)	P(losing year)	95th pct max DD
As in the post-freeze record (2023–26)	+0.384	1.46	+20R	–15R	+63R	18%	27R
As in confirmation (2019–22)	+0.587	1.93	+31R	–3R	+74R	7%	19R
Planning case: edge roughly halved	+0.190	1.16	+9R	–25R	+52R	34%	35R
Weak edge	+0.080	1.04	+3R	–32R	+46R	45%	40R
Edge gone	+0.000	0.96	–1R	–36R	+41R	53%	44R

Each row resamples the 2019–2026 trades (56 trades per simulated year) with the average trade shifted to the scenario value. In dollars, multiply R by about \$825 for 1 NQ or \$83 for 1 MNQ at 2026 volatility. The highlighted row is the planning case this report recommends.

Our planning case is the highlighted row: about +0.19R per trade, half the post-freeze average. That gives a median year of about +9R and roughly a 35% chance of a losing year at about 56 trades a year. A year like 2013 (–40R) remains possible.

What would make us more confident: 80 or more forward trades (about 1½–2 years) at +0.15R or better, with the runner days still showing up. **What would make us stop:** a drawdown beyond 35R (the planning-case 95th percentile), or 80 forward trades averaging below 0R.

Fixed risk on MNQ instead of fixed contracts

Bottom line. Fixed risk works for H25 short at any budget from \$150 up. In 2026 one MNQ risked about \$86 (median stop 43 points), so nothing is skipped from \$150 up. Only \$100 loses trades: 28% of 2026 signals. The profit factor barely changes: 1.61 on 1 NQ and about 1.57 at fixed risk after the freeze. Its real value to you is as a **third, uncorrelated sleeve** next to H41 and H47. It modestly raises an evaluation's pass rate and shortens the time to pass. Run it at the **low end** (\$100–\$150), because its evidence is the weakest of the three.

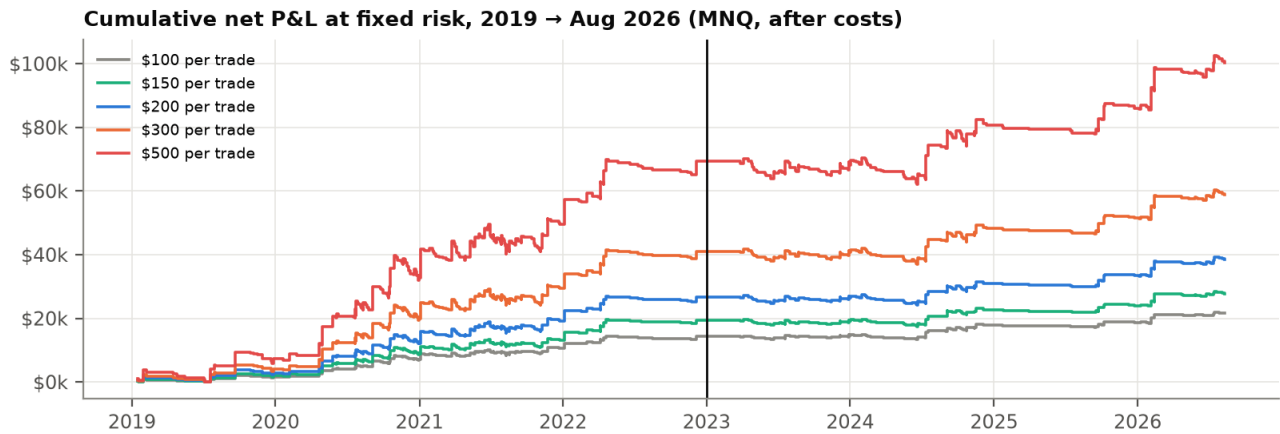
What changes

- **Sizing rule.** Contracts = risk budget ÷ (stop distance in points × \$2), rounded down; no trade if 0. MNQ costs are assumed at \$2.50 per round trip (\$4.25 stress).
- **Profit factor roughly unchanged.** After the freeze, trades on the calm and middle thirds of days averaged +0.45R and +0.52R, and the most volatile third +0.18R. Fixed contracts put the most dollars on the volatile days, fixed risk does not. The two effects roughly cancel, so PF moves from about 1.6 to about 1.55.
- **The 25% partial needs at least 3 MNQ.** The partial is rounded to the nearest whole contract (ties down), so at 1–2 MNQ everything runs to the close. That happens on most 2026 trades at \$100–\$150. Historically the runner leg alone averaged +0.66R, compared with +0.51R for the frozen plan. That is descriptive only.
- **Losses come in clusters.** It can fire up to 4–5 times on the same morning when NQ breaks away from ES. At \$200 per trade the worst post-freeze day was –\$557 and the worst drawdown \$3,196. Check this against your daily loss limit.

Post-freeze 2023 → Aug 2026 (data never used to select the strategy)

Metric	1 NQ fixed	1 MNQ fixed	\$100 risk	\$150 risk	\$200 risk	\$300 risk	\$500 risk
Trades taken (skipped)	169 (0)	169 (0)	160 (9)	169 (0)	169 (0)	169 (0)	169 (0)
Profit factor	1.61	1.61	1.74	1.55	1.57	1.54	1.55
Profit factor, stress costs	1.56	1.55	1.65	1.46	1.48	1.45	1.47
Win rate	20%	19%	20%	19%	20%	19%	19%
Net P&L	\$33,173	\$4,239	\$7,348	\$8,403	\$11,922	\$17,810	\$30,976
Net P&L per year	\$9,189	\$1,174	\$2,035	\$2,328	\$3,303	\$4,934	\$8,581
Average trade	\$196	\$25	\$46	\$50	\$71	\$105	\$183
Average \$ risk per trade	\$476	\$48	\$79	\$126	\$175	\$276	\$475
Risk range (min – max)	\$130 – \$1,480	\$13 – \$148	\$50 – \$100	\$77 – \$150	\$101 – \$200	\$202 – \$300	\$404 – \$500
Contracts (avg / max)	1.0 / 1	1.0 / 1	2.1 / 7	3.3 / 11	4.5 / 15	7.1 / 23	12.1 / 38
Trades that can take the 25% partial	100%	0%	28%	68%	86%	95%	100%
Max drawdown (closed trades)	\$6,188	\$806	\$1,590	\$2,314	\$3,196	\$4,964	\$8,358
Worst losing run	–\$5,329	–\$658	–\$1,002	–\$1,886	–\$2,310	–\$4,013	–\$6,681
Worst day	–\$2,453	–\$248	–\$270	–\$437	–\$557	–\$945	–\$1,517

1 NQ = the frozen strategy exactly. MNQ columns use \$2/point, \$2.50 per round trip per contract, and the partial rounded to whole contracts. Drawdowns use closed-trade equity.



Full out-of-sample record 2019 → Aug 2026

Metric	1 NQ fixed	1 MNQ fixed	\$100 risk	\$150 risk	\$200 risk	\$300 risk	\$500 risk
Trades taken (skipped)	424 (0)	424 (0)	415 (9)	424 (0)	424 (0)	424 (0)	424 (0)
Profit factor	1.81	1.80	1.85	1.71	1.72	1.71	1.71
Profit factor, stress costs	1.74	1.72	1.72	1.60	1.60	1.59	1.60
Win rate	21%	20%	20%	20%	20%	20%	20%
Net P&L	\$90,282	\$11,434	\$21,629	\$27,715	\$38,518	\$58,838	\$100,240
Net P&L per year	\$11,864	\$1,502	\$2,842	\$3,642	\$5,061	\$7,732	\$13,172
Average trade	\$213	\$27	\$52	\$65	\$91	\$139	\$236
Average \$ risk per trade	\$398	\$40	\$82	\$130	\$180	\$281	\$480
Risk range (min - max)	\$65 - \$1,480	\$6 - \$148	\$50 - \$100	\$77 - \$150	\$101 - \$200	\$202 - \$300	\$404 - \$500
Contracts (avg / max)	1.0 / 1	1.0 / 1	2.7 / 15	4.2 / 23	5.8 / 30	8.9 / 46	15.1 / 76
Trades that can take the 25% partial	100%	0%	42%	79%	92%	98%	100%
Max drawdown (closed trades)	\$6,188	\$806	\$1,689	\$2,562	\$3,497	\$5,132	\$9,265
Worst losing run	-\$5,329	-\$658	-\$1,144	-\$1,886	-\$2,310	-\$4,013	-\$6,681
Worst day	-\$2,453	-\$248	-\$400	-\$685	-\$858	-\$1,321	-\$2,184

Stop size and contracts by year

Year	Trades	Median stop (pts)	1 MNQ risk	\$100: MNQ (skip)	\$150: MNQ (skip)	\$200: MNQ (skip)	\$300: MNQ (skip)	\$500: MNQ (skip)
2019	25	7.0	\$14	7 (0%)	10 (0%)	14 (0%)	21 (0%)	35 (0%)
2020	81	17.5	\$35	2 (0%)	4 (0%)	5 (0%)	8 (0%)	14 (0%)
2021	122	16.0	\$32	3 (0%)	4 (0%)	6 (0%)	9 (0%)	15 (0%)
2022	27	22.2	\$44	2 (0%)	3 (0%)	4 (0%)	6 (0%)	11 (0%)
2023	53	17.8	\$36	2 (0%)	4 (0%)	5 (0%)	8 (0%)	14 (0%)
2024	66	21.0	\$42	2 (0%)	3 (0%)	4 (0%)	7 (0%)	12 (0%)
2025	18	20.1	\$40	2 (0%)	3 (0%)	4 (0%)	7 (0%)	12 (0%)
2026	32	43.1	\$86	1 (28%)	1 (0%)	2 (0%)	3 (0%)	5 (0%)

MNQ = median contracts per trade at that budget; skip = share of signals where 1 MNQ risks more than the budget.

Prop-firm evaluation

Each simulated evaluation draws random whole trading days from 2019-Aug 2026 (every trade of that day, in time order) for up to 250 trading days. It passes at the profit target and fails at the trailing drawdown (closed trades) or the daily loss limit. 5,000 runs per row. The rules are generic examples. The last three rows add H25 to an account that already runs H41 (at \$150, 1 MNQ up to \$250) and H47 (at \$150).

Account set-up	Pass	Fail	Not done in 250 days	Median days to pass
Example A: target \$3,000 · trailing DD \$2,000 · daily loss \$1,000				
H25 alone, \$150 per trade	60%	21%	19%	117
H25 alone, \$250 per trade	56%	44%	0%	56
H25 alone, \$500 per trade	40%	60%	0%	17
H41 + H47 at \$150 each (reference, no H25)	58%	42%	0%	38

Account set-up	Pass	Fail	Not done in 250 days	Median days to pass
H41 + H47 + H25, all at \$150	58%	42%	0%	28
H41 + H47 at \$150, H25 at \$100	60%	40%	0%	31
Example B: target \$6,000 · trailing DD \$3,000 · daily loss \$2,000				
H25 alone, \$150 per trade	24%	6%	70%	180
H25 alone, \$250 per trade	51%	33%	16%	124
H25 alone, \$500 per trade	43%	57%	0%	44
H41 + H47 at \$150 each (reference, no H25)	63%	35%	2%	95
H41 + H47 + H25, all at \$150	66%	34%	0%	73
H41 + H47 at \$150, H25 at \$100	67%	32%	1%	80

- **Alone, H25 is slow and fragile.** At \$150, 19% of Example A runs and 70% of Example B runs had not finished after 250 trading days. Larger risk speeds it up but fails more often. With a 19% win rate, losing streaks of 14 trades have happened after the freeze. It is not a strategy to take through an evaluation on its own.
- **As a third sleeve it helps a little, mainly with speed.** H41 + H47: Example A 58% pass in a median 38 days; Example B 63% in 95 days. Adding H25 at \$100: 60% in 31 days, and 67% in 80 days. Its daily correlation with H41 is about 0.14 and with H47 about 0.
- **Suggested use:** add it last, at \$100–\$150, only once H41 and H47 are running. Given its weaker evidence, keep it smaller than the other two until its own forward record supports it.
- Many firms trail the drawdown on unrealised equity, and H25 holds its runners to 15:59. The day-resampling also does not capture long bad regimes. Real pass rates will be somewhat lower.

Discussion

Strength of evidence. Over 2010–2026: +266R on 897 trades (+0.30R per trade, t 2.88 in R, PF 1.68 in dollars). The post-freeze part is 169 trades at +0.38R (t 1.66). Of the three candidates reported, this has the weakest untouched evidence but the best search-corrected confirmation.

How to read the dollar curve. NQ's price and volatility rose sharply, so dollar risk per trade on 1 NQ grew from about \$50 (2010) to about \$825 (2026). Most dollar P&L is therefore post-2019. In R, the picture is flatter: modest gains 2011–12, losses 2013 and 2015, and steady gains from 2017.

Where the P&L comes from. All of it comes from the fifth of trades still open at 16:00. Entries in the first hour averaged +0.40R (+0.51R after the freeze); later entries were roughly flat or negative. After the freeze, the most volatile third of days contributed least (+0.18R per trade). This is descriptive only.

Management. The frozen 25%/75% plan and the plain hold-to-close exit worked, and so did wide trailing stops. Nearly everything that takes profit early lost or barely broke even. Confirmation had already flagged this (management support 0.15). Treat it as a feature of the strategy: its exits must never be tightened.

Portfolio fit. The three strategies are close to independent. H41 is all-day momentum in either direction, H47 is a long-only gap-and-go at 09:45, and H25 short is Nasdaq-specific weakness, mostly in the first hour. Its value is diversification, not standalone return.

Recommendation

Recommendation: THIRD CANDIDATE, LOWER CONVICTION. Add it last, at small risk, as a third sleeve next to H41 and H47. Do not trade it alone. It was profitable in every research stage, and both post-freeze tests were classified SUPPORTED (holdout +\$17.6k, PF 1.47; 2026 +\$15.6k, PF 1.93). Its daily P&L is essentially uncorrelated with H41 and H47. But the untouched holdout was only marginal (t 1.19, null p 0.059). It needs its exact stop/exit plan, it wins only 19% of the time, and a handful of trades carry the result. It is a clear step down from the first two candidates.

Decision requested: approve H25 short as the **third, lower-conviction sleeve**, added only after H41 and H47 are running, at the smallest risk of the three.

Pilot design. MNQ, fixed risk \$100–\$150 per trade: contracts = risk $\div$ (ATR_h $\times$ \$2), rounded down. The partial is rounded to whole contracts, so at 1–2 MNQ everything runs to the close. Keep the exits exactly as frozen. Because of the morning clusters, check that 4–5 stop-outs in one session stay inside the daily loss limit.

Alternative: forward-paper only via forward/run_forward.py until the gate below is met. Given its weaker evidence, this is a reasonable choice.

Conditions, limits and kill criteria

- Scale-up gate: at least 80 true-forward trades averaging +0.15R or better, net positive under stress costs, and no drawdown beyond 22R (the bootstrap 95th percentile at the observed edge). Only then raise it to the same risk as H41 and H47.
- Kill switch: a drawdown beyond 35R (the planning-case 95th percentile), or 80 forward trades averaging below 0R.
- Exit discipline: never add targets or early break-even rules. Across all periods, those are the variants that mostly lost money.
- Keep H25 short and its sibling, cohort member #14 (H25 both directions), in one budget. They share the same signal.
- No parameter, filter or management changes. Every observation in this report is descriptive.

Caveats

- Backtests use 1-minute bars with pessimistic same-bar handling and fixed slippage. They do not model queue position, fast-market slippage beyond 2 ticks per side (stress case), or exchange outages.
- The 2026 sample (Jan–10 Aug) is short. The campaign's own record rates it low-power (cohort null p 0.26).
- The R-based view and the parameter/management grids in this report were computed after the holdout and 2026 results were known. They are diagnostic only and must not be used to modify or re-select the

strategy. The preregistered process says the only valid new evidence now comes from the true-forward record starting 24 Sep 2026.

- This is a quantitative research assessment of a systematic strategy, not personalised investment advice. Capital allocation decisions rest with the committee.

Appendix A: frozen specification (verbatim)

H25 (#21, short only)

```
{
  "costs": {
    "baseline_round_trip_usd": 14.0,
    "cost_version": "costs_v2_approved_2026-09-21#a27ed5df0c04"
  },
  "direction": "short",
  "entry_conditions": {
    "down_event": "30-minute NQ/ES correlation <= 0.65 and NQ's 30-minute beta residual <= -2.0 sigma",
    "mode": "continuation",
    "rule": "continuation: BUY on up-event, SELL on down-event; direction=short",
    "up_event": "30-minute NQ/ES correlation <= 0.65 and NQ's 30-minute beta residual >= +2.0 sigma"
  },
  "entry_id": "E7cd08682905f49b9",
  "family": "v3_h25_corr_breakdown",
  "instrument": "NQ",
  "management": {
    "abs_target": null,
    "be_R": null,
    "be_after_leg": null,
    "exit_opp": false,
    "hold": "eod",
    "id": "P25_1R_RUN",
    "kind": "partial",
    "legs": [
      [
        0.25,
        1.0
      ],
      [
        0.75,
        null
      ]
    ],
    "needs_stop": true,
    "params": {},
    "trail": null
  },
  "parameters": {
    "cth": 0.65,
    "mode": "continuation",
    "rz": 2.0
  },
  "stop": {
    "method": "atr",
    "mult": 1.0
  },
  "strategy_id": "C2845a074bb5a5501",
  "trading_window": {
    "entry_end": "15:30",
    "entry_start": "09:30",
    "flat_at": "16:00",
    "name": "rth"
  }
}
```

Appendix B: provenance

Item	Value
Final cohort freeze	freezes/FINAL_COHORT_FREEZE.json · sha256 f5666dac728aa2539b2f8223309012914d71923ab71f3363e9e035344ace8f03
Frozen (UTC)	2026-09-22T23:21:51.429326+00:00
Discovery freeze	sha256 0343a974953d53dc026fd1f7ef5c124c8985213b132fba5df4f8a986fc85b2c9
Master record	results/V3_MASTER_RESEARCH_RECORD.md
Sealed holdout / audit trades	results/holdout/trades.parquet · results/audit2026/trades.parquet
This report	reports/H25_Correlation_Breakdown_Short_Research_Report.pdf, built from trades regenerated with the frozen specs (exact match with the sealed records) + stored discovery/confirmation evidence
Known erratum	results/audit2026/ERRATUM_2026-09-22.json (X3 null p only; does not affect this report)